



East West University

Lab Report

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Course Title: Electronic Circuits **Course Code:** CSE251

Sec: 04

Expt No: 01

Expt Name: I-V Characteristics and Modeling of Forward Conduction of a Diode.

Group No: 06

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Title: I-V Characteristics and Modeling of Forward Conduction of a Diode.

Objectives:

1. To measure the I-V characteristics of forward conduction of a p-n junction diode.
2. To determine the models of forward conduction of a p-n junction diode.

Introduction:

Diode is one of the most basic nonlinear electronic devices. An ideal diode acts like a switch for electric current, acting as a short circuit for current flow in one direction (forward bias connection) while behaving as an open circuit for current flow in the opposite direction (reverse bias connection). The characteristics of practical diodes are somewhat different from those of ideal ones. The p-n junction diodes are one of the most popular diodes used in the industry. The forward bias current-voltage (I-V) characteristic of a p-n junction diode will be measured in this experiment.

Circuit Diagram:

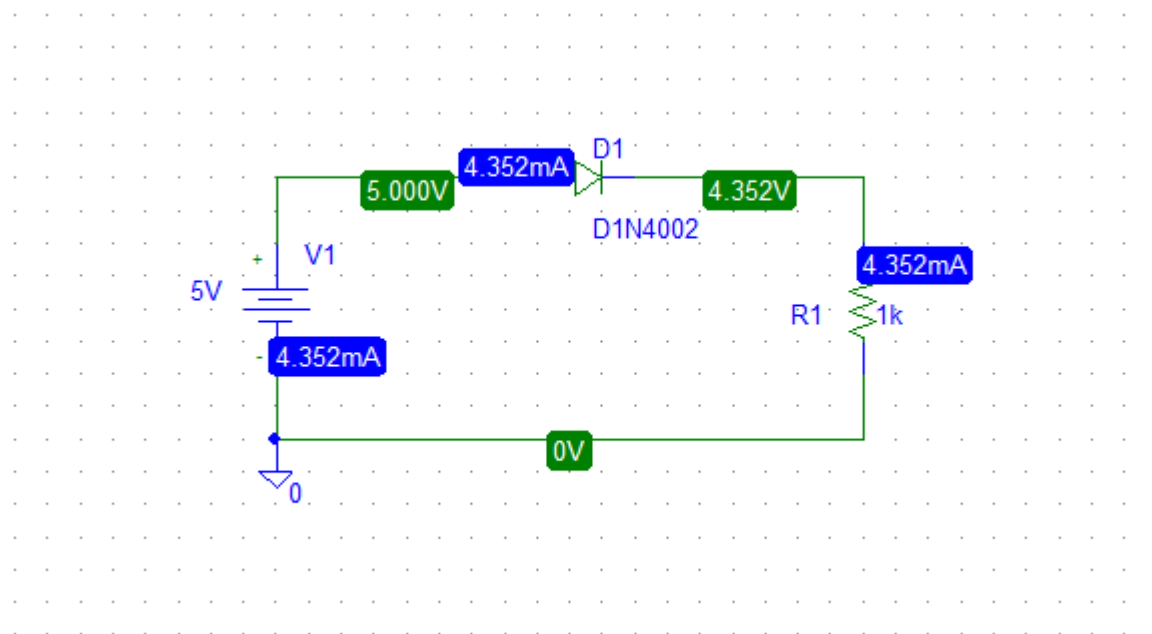


Figure-1: Circuit set up to measure forward bias I-V characteristics of a diode

Table 1. Experimental Datasheet:

V_s (V)	V_D (V)	V_R (V)	I_D (mA)=V_R/R(KΩ)	Measured Value of R(KΩ)
0.5	0.34	0.016	0.01481	
1	0.18	0.048	0.04444	
1.5	0.49	0.74	0.68519	
2	0.51	1.16	1.07407	
2.5	0.52	1.9	1.75926	
3	0.29	1.94	1.79630	
3.5	0.29	3.2	2.96296	
4	0.31	3.7	3.42593	
4.5	0.31	4.2	3.88889	
5	0.32	4.8	4.44444	
5.5	0.33	5.3	4.90741	
6	0.33	5.8	5.37037	
6.5	0.34	6.4	5.92593	
7	0.34	6.9	6.38889	
7.5	0.34	7.4	6.85185	
8	0.35	7.94	7.35185	
8.5	0.35	8.53	7.90741	
9	0.36	7.1	6.57407	1.08 K
9.5	0.36	9.58	8.87037	
10	0.36	10.0	9.25926	
10.5	0.37	10.5	9.72222	
11	0.37	11.11	10.28704	
11.5	0.37	11.63	10.76389	
12	0.38	12.2	11.29630	
12.5	0.38	12.75	11.80556	
13	0.39	13.25	12.26852	
13.5	0.39	13.74	12.72222	
14	0.39	14.32	13.25926	
14.5	0.40	14.75	13.65741	
15	0.404	15.27	14.13889	
15.5	0.407	15.8	14.62963	
16	0.410	16.4	15.18519	
16.5	0.413	16.77	15.53704	
17	0.417	17.45	16.15741	
17.5	0.419	18.0	16.66667	

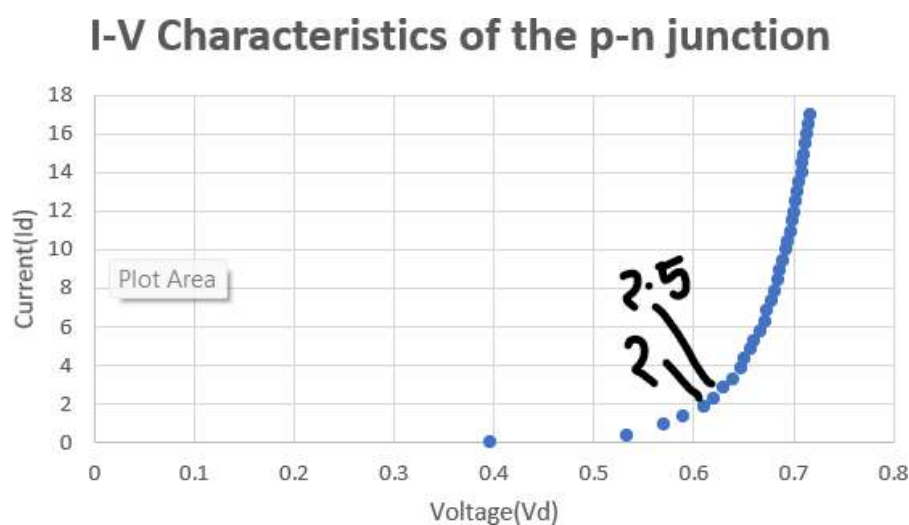
Post-Lab Report Questions and Solutions:

1. Using MATLAB, plot the I-V characteristics of the p-n junction diode in forward conduction. Label the axes appropriately and have it printed.

2. Use pencil to identify the points on your graph that are corresponding to $I_D = 2 \text{ mA}$ and $I_D = 2.5 \text{ mA}$. Use these data points to calculate the diode parameters I_S and n from the equation

$$I_D = I_S \exp [V_D / nV_T] . \text{ Use } V_T = 0.0259 \text{ V}$$

Solve:



When, $I_{D1} = 2 \text{ mA}$ then $V_{D1} = 0.52 \text{ V}$

And $I_{D2} = 2.5 \text{ mA}$ then $V_{D2} = 0.53 \text{ V}$

Now to find n we can use this equation,

$$V_{D2} - V_{D1} = nV_T \ln\left(\frac{I_{D2}}{I_{D1}}\right)$$

$$\Rightarrow n = (V_{D2} - V_{D1}) / V_T \ln(I_{D2} / I_{D1})$$

$$\Rightarrow n = (0.53 - 0.52) / 0.0259 \ln(2.5 / 2)$$

$$\Rightarrow n = 1.73$$

And,

$$I_D = I_S e^{[V_D / nV_T]}$$

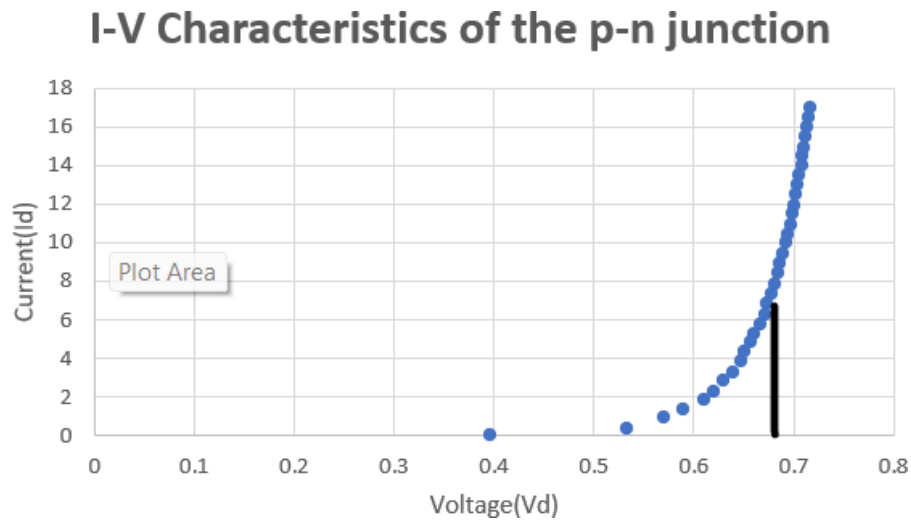
$$\Rightarrow I_S = I_D / e^{[V_D / nV_T]}$$

$$\Rightarrow I_S = 2 / e^{[0.52 / 1.73 \times 0.0259]}$$

$$\Rightarrow I_S = 0.0182 \text{ mA}$$

3. Determine the cut-in voltage from the printed graph by drawing the extrapolated line with the pencil.

Solve:



From the above graph, we can see, the cut-in voltage is around 0.682

4. If the diode resistance for the piecewise linear model is defined as

$1/r_D = \partial I_D / \partial V_D = (I_{D2} - I_{D1}) / (V_{D2} - V_{D1})$, calculate the value of R_D from the data points corresponding to $I_D = 2\text{mA}$ and $I_D = 2.5\text{mA}$

Solve:

Here,

$$I_{D1} = 2 \text{ mA}, V_{D1} = 0.615 \text{ V and, } I_{D2} = 2.5\text{mA}, V_{D2} = 0.626\text{V}$$

We know,

$$1/R_D = (I_{D2} - I_{D1}) / (V_{D2} - V_{D1})$$

$$R_D = (V_{D2} - V_{D1}) / (I_{D2} - I_{D1})$$

$$= (0.626 - 0.615) / (2.5 - 2)$$

$$= 0.022 \text{ K}\Omega$$

So, $R_D = 0.022 \text{ K}\Omega$

5. Simulate the circuit of Figure 1 for a DC bias (V_S) range of 0-5 volts using PSpice.

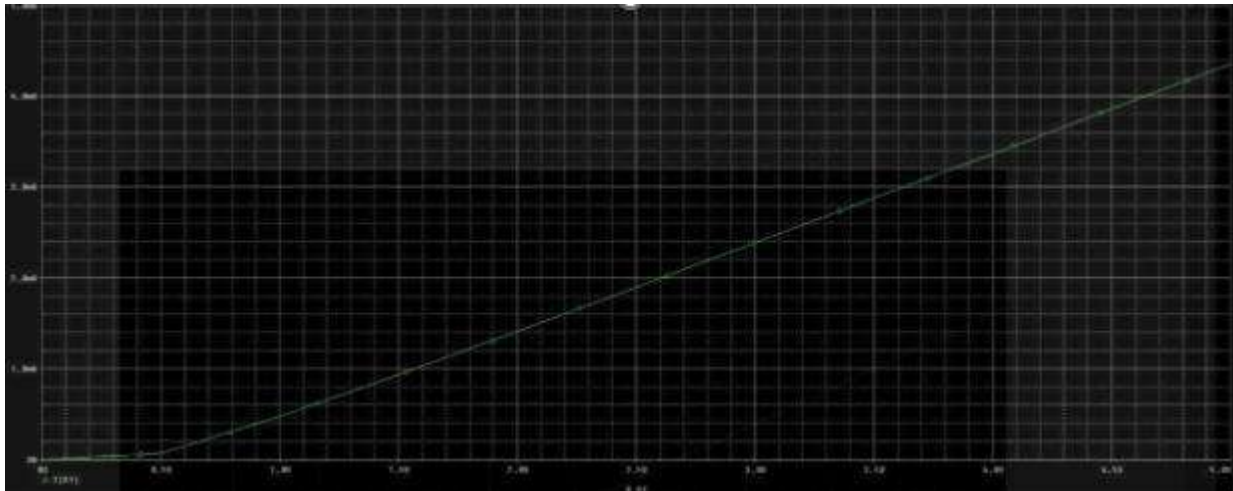


Figure 2: V_S vs I_D Graph

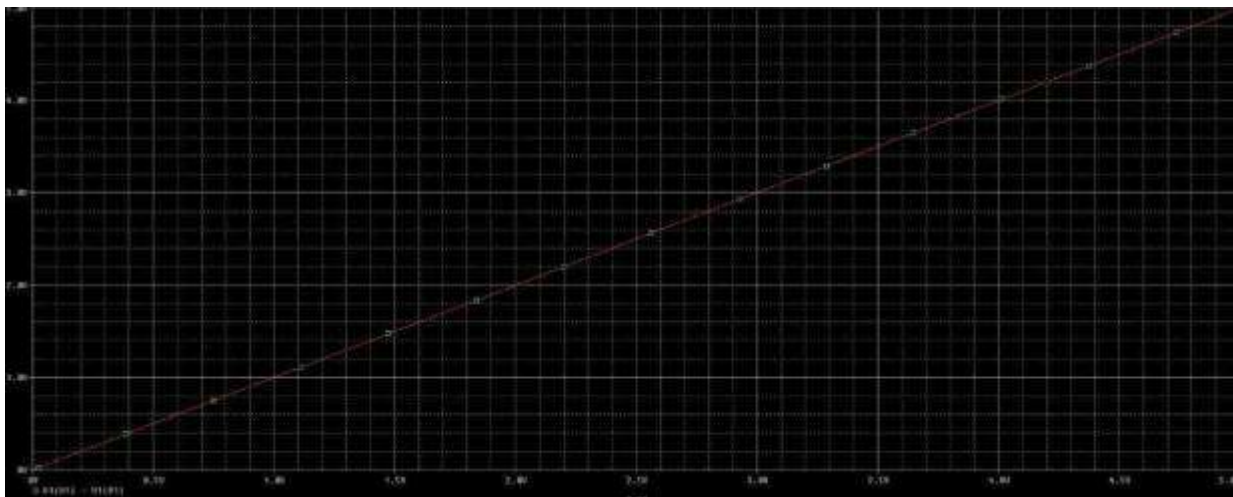


Figure 3: V_S vs V_D Graph

Discussion:

This experiment successfully examined the I-V characteristics of a p-n junction diode in forward conduction. The observed results closely matched the expected exponential behavior.

The experiment data and I-V characteristics of the p-n junction curve graph clearly illustrate the typical forward bias behaviour of a diode. Initially at low (approximately 0.5V) forward voltages (V_0), the current (I_0) through the diode was very small, indicating that the diode was in its non-conducting state. As the forward voltage increased, the diode began to conduct significantly, and the current increased rapidly after a certain cut-in voltage. The theoretical model was confirmed by the exponential rise in the I-V curve.

From the graphical analysis, the cut in voltage was 0.682 V which is consistent with the typical cut-in voltage for silicon diodes (0.7 V). The cut-in voltage represents the approximate voltage at which the diode begins to conduct significantly.

From the Shockley diode equation ($I_0 = I_s \exp[V_0/nV_T]$), the calculated n was 1.73, which is within the range (1 to 2 for silicon diodes), indicating that the diode operates close to ideal conditions. Subsequently, the reverse ^{saturation} current (I_s) was 0.0182 mA, which is relatively high.

Overall, the experiment successfully characterized the forward conduction of a p-n junction diode with its I-V characteristics. The derived parameters provided a quantitative understanding of the diode's intrinsic properties. The extracted parameters can be useful in analyzing and designing circuits that utilize diodes.

Future experiments could explore the temperature dependence of diode characteristics and investigate the reverse bias region to gain a complete understanding of diode behavior.